SAMUEL AKINWUMI ADENIJI

AI for Healthcare · Multimodal Medical AI · Medical Image Analysis

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RESEARCH INTERESTS

Medical vision–language models for 3D imaging; instruction-conditioned and token-efficient architectures for volumetric data; grounded radiology report generation and multi-scale representation learning; interpretable and reliable medical AI; deployment of imaging AI in low-resource and African clinical settings.

EDUCATION

Carnegie Mellon University — Master of Science, Engineering Artificial Intelligence *Aug 2025 – May 2027*

Concentration: AI in Healthcare (Research) · *Grade point average: 3.94 / 4.00*

Federal University of Technology, Minna, Nigeria — Bachelor of Engineering, Mechatronics Engineering *Sep 2018 – Jan 2025*

First Class Honors · *Cumulative grade point average: 4.60 / 5.00* · *Concentration: Embedded Systems and Machine Learning*

RESEARCH AND TEACHING EXPERIENCE

Research Intern — Language and Spatial Laboratories *June 2026 – Present*

Carnegie Mellon University · Supervisor: Prof. Prasenjit Mitra

- Research on vision–language models for 3D medical imaging; lead developer of ICTC (see Research Projects).
- Built the full research pipeline, from data ingestion and the compression projector to multi-task training and evaluation across six visual token budgets.

Research Assistant (Intern) *Oct 2023 – Jan 2025*

Artificial Intelligence for Clean Energy Limited, Minna, Nigeria · Supervisor: Dr. Jibril Abdullahi Bala

- Built a Flask and Random Forest energy-monitoring platform on Google Cloud (99% uptime), improving monitoring accuracy by 30%.
- Piloted drones for field data collection, cutting collection time by 30% and improving data quality by 12%.

Teaching Assistant — AI System Design (04-652) *Fall 2026*

Carnegie Mellon University Africa · Supervisor: Prof. Hatem Haddad

- Supporting a course on edge-to-cloud AI systems, from embedded machine learning on Raspberry Pi and Arduino to cloud-based model analysis.

Teaching Assistant — Cloud Computing (15-319 / 15-619) *Spring 2026*

Carnegie Mellon University · Supervisor: Prof. Majd Sakr

- Supported more than 200 undergraduate, master's and doctoral students through office hours, course-overview sessions and grading.

Peer Instructor (Volunteer) — Introduction to Machine Learning *Spring 2026*

Carnegie Mellon University

- Led small-group tutoring on regression, classification, model evaluation and the underlying linear algebra and probability.

RESEARCH PROJECTS

ICTC: Instruction-Conditioned Token Compression for 3D CT Vision–Language Models *June 2026 – Present*

Language and Spatial Laboratories, Carnegie Mellon University · Advisor: Prof. Prasenjit Mitra

Stack: Python, PyTorch, Vision Transformers, Perceiver, LoRA, Hugging Face, CT-RATE

- Designed a two-stage token compressor for 3D chest CT report generation. A Perceiver stage condenses each slice to 32 latents, then an instruction-conditioned stage condenses the slices into a fixed budget of visual tokens for a LoRA-adapted language model.
- On the CT-RATE validation set (3,039 scans): ROUGE-L 0.273, BLEU-4 0.128 and RaTEScore 0.801 for full report generation.
- Built a multi-task instruction pipeline (report generation, entity questions, abnormality question answering) with a 368-test suite, evaluated across six token budgets from 16 to 512.
- Now adding instruction-conditioned plane selection, where the clinical question chooses an interpretable, anatomy-aware 3D viewing plane.

FPN-Mamba: Linear-Time Bidirectional State Space Feature Pyramids for Small-Lesion Brain MRI Classification *Feb 2026 – Present*

Advisor: Prof. Moise Busogi

Stack: Python, PyTorch, Feature Pyramid Networks, Mamba state-space models, scikit-learn

- Combined a feature pyramid network with bidirectional Mamba state-space blocks to detect small brain metastases in MRI, in linear time.
- Against a re-implemented InceptentionNet (5 seeds $\times$ 3 folds), reaches AUC 0.982 vs 0.897 and smallest-lesion recall 0.870 vs 0.788 (all $p < 0.02$), and trains 3.7$\times$ faster per step than a matched self-attention design at 512 pixels.
- Accepted at the Africa in AI Workshop, NeurIPS 2026 — OpenReview.

Investigating White Blood Cells as a Source of False-Positive Malaria Parasite Detection in African Blood-Smear Images *Jan 2026 – Aug 2026*

Stack: Python, PyTorch, Ultralytics YOLOv12, OpenCV, SciPy, DBSCAN, Weights & Biases

- Ran a controlled study on 8,000 Giemsa-stained thick blood-smear images from Uganda and Ghana, comparing detectors trained with and without white blood cell labels.
- Seven independent spatial and statistical analyses (including Ripley's Cross-K, DBSCAN and Mann–Whitney U) refuted the common belief that white blood cells cause false-positive parasite detections, pointing to staining artefacts instead.
- Accepted for oral presentation at the MIRASOL Workshop, MICCAI 2026 — paper · arXiv:2609.29663.

Condition-Aware Intensive Care Unit Early-Warning System *Jan 2026 – May 2026*

Stack: Python, PyTorch, TensorFlow, MIMIC-IV, FastAPI, Docker, WebSockets, scikit-learn

- Benchmarked six diagnosis-conditioned sequence models (LSTM, GRU, TCN, BiLSTM with attention, Transformer, TFT-style) for deterioration risk on MIMIC-IV intensive care data; the best, a BiLSTM with attention, reaches ROC-AUC 0.76.
- Deployed it as a Dockerized FastAPI service with WebSocket streaming inference, built for reproducibility under PhysioNet data governance.

PUBLICATIONS

1. **S. A. Adeniji**, G. C. Obasi, C.-V. Ntwali, A. M. Iorumbur, C. Raymond, L. Uwimana, A. T. Issah. "Investigating White Blood Cells as a Source of False-Positive Malaria Parasite Detection in African Blood-Smear Images." *Medical Image Computing in Resource Constrained Settings Workshop & Knowledge Interchange (MIRASOL), International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2026*, Lecture Notes in Computer Science, vol. 17264, Springer (oral presentation). [MICCAI] · [arXiv] · [PDF] · [OpenReview]

2. S. D. Adetoro, **S. A. Adeniji**, M. Busogi. "FPN-Mamba: Linear-Time Bidirectional State Space Feature Pyramids for Small-Lesion Brain MRI Classification." *Africa in AI Workshop, Conference on Neural Information Processing Systems (NeurIPS) 2026.* [OpenReview]

3. K. E. Jack, **S. A. Adeniji**. "Convergence of Blockchain, Smart Microgrid and Energy Market." *Electrica*, 24(2), 367–374, 2024. DOI: 10.5152/electrica.2024.23132.

4. K. E. Jack, **S. A. Adeniji**. "Development of an Artificial Intelligence-Based Infant's Emotion Recognition System for a Caregiving Robot." *Nigerian Journal of Engineering and Applied Sciences*, 2024. DOI: 10.5455/NJEAS.209325.

ACADEMIC SERVICE AND RESEARCH TRAINING

- **SPRINT / SPARK AI for African Medical Imaging Knowledge Translation**, 2026 Cohort — state-of-the-art methods for glioma brain-tumour segmentation and translation of imaging AI into African clinical practice; led a capstone on malaria parasite detection.

TECHNICAL SKILLS

- **Languages and Platforms:** Python, Linux/Bash, Git, LaTeX, Microsoft Azure, Google Cloud Platform.
- **Deep Learning:** PyTorch, TensorFlow, Vision Transformers, Mamba / state-space models, CNNs and Feature Pyramid Networks, vision–language models, LoRA / parameter-efficient fine-tuning, HuggingFace.
- **Medical Imaging and Data:** 3D CT and MRI pipelines, CT-RATE, MIMIC-IV, MONAI-style preprocessing, OpenCV, YOLO, scikit-learn, Weights & Biases, Docker.
- **Foundations:** Linear algebra, probability and statistics, mathematical modelling, experimental design.

AWARDS AND SCHOLARSHIPS

- **Mastercard Foundation Scholarship**, Carnegie Mellon University (2025), for the master's in Engineering Artificial Intelligence.
- **MINDS Scholarship** (Mandela Institute of Developmental Studies), Carnegie Mellon University (2026).
- **Best Graduating Student Project**, Department of Mechatronics Engineering, Federal University of Technology, Minna (2024): a nanny robot for childcare and monitoring.